

Surface Pattern Recognition using Shape Subspace

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Abstract

This progress report presents significant advancements in surface anomaly detection for point cloud data, building upon our previously established Shape Subspace methodology. While our initial work focused on fixed patch analysis using synthetic data, we have now extended the approach to include a sliding window technique and validated it on real-world PLY files. The sliding window method introduces overlapping analysis regions with configurable stride, providing $3.5\times$ higher spatial resolution compared to fixed patches. Experimental validation on real PLY data (61,914 points) demonstrates that the sliding window approach achieves superior visual contrast for anomaly detection, with improved signal-to-background ratio despite lower absolute peak magnitudes. The key finding is that method selection should be data-dependent: fixed patches perform better on clean synthetic data, while sliding windows excel with noisy real-world data. This work establishes practical guidelines for surface anomaly detection method selection and demonstrates successful transition from synthetic to real-world applications.

1 Background and Previous Work

1.1 Previous Work Summary

Our initial research established a framework for surface pattern recognition on synthetic three-dimensional floor point clouds using Shape Subspace methods combined with Difference Subspace (DS) techniques. The methodology involved segmenting synthetic floor surfaces (50×50 units, 10,000 points) into uniform patches, generating shape subspaces via Singular Value Decomposition (SVD), and computing difference subspaces between neighboring patches to detect local surface variations.

Key findings from our previous work demonstrated that:

- Smaller patch sizes (3×3 , 4×4) provided higher accuracy for fine-grained surface detection
- Four-neighbor configurations delivered better sensitivity to local variations
- Eight-neighbor approaches offered improved stability for complex patterns
- The method successfully detected various synthetic patterns (bumps, ridges, corners)

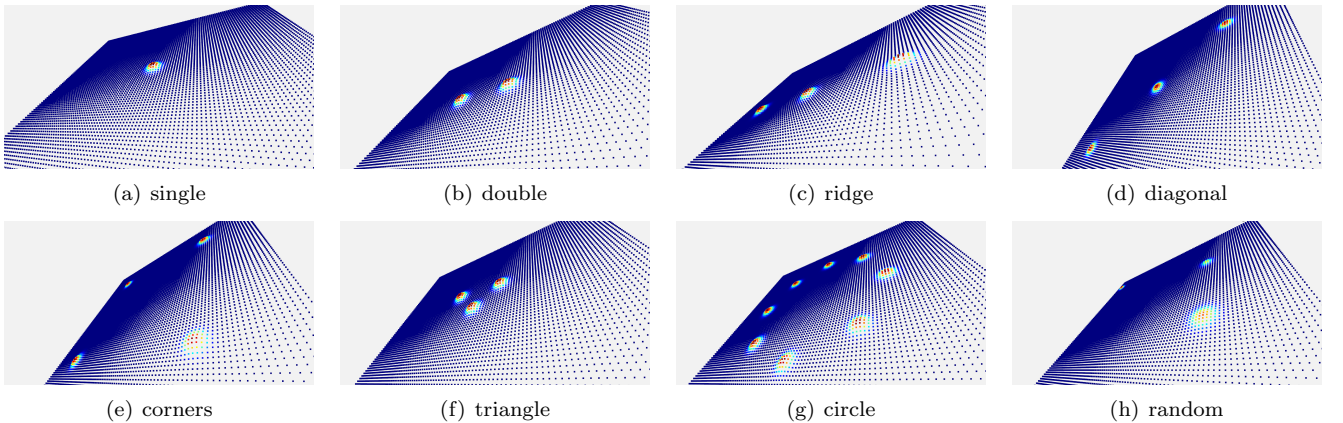


Figure 1: Generated patterns

1.2 Motivation and Objectives

Although our initial approach proved effective on synthetic data, limitations became apparent: (1) boundary sensitivity at patch edges, (2) limited spatial resolution from non-overlapping patches, and (3) unvalidated performance on real-world noisy data.

This progress report addresses these through the development of a sliding window methodology with overlapping analysis regions, the validation of real PLY files (61,914 points), and the establishment of data-dependent method selection guidelines. The sliding window approach provides continuous coverage without rigid boundaries, bridging the gap between synthetic validation and real-world application.

2 Extension to Sliding Window Approach

2.1 Sliding Window Methodology

The sliding window approach extends our fixed patch methodology by introducing overlapping analysis regions that move across the point cloud surface with configurable stride parameters. This addresses the boundary sensitivity limitations of fixed patches while maintaining the mathematical rigor of the shape subspace framework.

2.1.1 Mathematical Formulation

Given a point cloud with box dimensions that limit (W, H) , the sliding window creates a grid of overlapping analysis windows. For a desired window size (w, h) and overlap percentage $\alpha \in [0, 1)$, the stride parameters are calculated as:

$$s_x = w \cdot (1 - \alpha) \quad (1)$$

$$s_y = h \cdot (1 - \alpha) \quad (2)$$

The resulting grid dimensions become:

$$N_x = \left\lfloor \frac{W - w}{s_x} \right\rfloor + 1 \quad (3)$$

$$N_y = \left\lfloor \frac{H - h}{s_y} \right\rfloor + 1 \quad (4)$$

Each window $W_{i,j}$ at grid position (i, j) encompasses points within the region:

$$\mathcal{R}_{i,j} = \{(x, y, z) \mid x_{min} + i \cdot s_x \leq x < x_{min} + i \cdot s_x + w, \quad (5)$$

$$y_{min} + j \cdot s_y \leq y < y_{min} + j \cdot s_y + h\} \quad (6)$$

2.2 Real-World Data Processing

For PLY files, we implement preprocessing (centering, optional downsampling), adaptive window sizing based on point density, boundary handling for edge cases, and minimum point filtering to ensure meaningful subspace computation. This enables robust analysis while maintaining theoretical compatibility.

3 Real-World Validation and Comparative Analysis

3.1 Experimental Setup

We compared both approaches on synthetic data (50×50 surfaces, 10,000 points, various patterns) and real PLY data (floor.ply, 61,914 points). Configuration: 8×8 grid, subspace dimension 3, eight-neighbor computation, 50% sliding window overlap.

3.2 Results

3.2.1 Synthetic Data Performance

On synthetic data, both approaches demonstrated effective pattern detection with comparable performance characteristics:

- **Fixed Patches:** Max magnitude = 0.652, Mean = 0.494

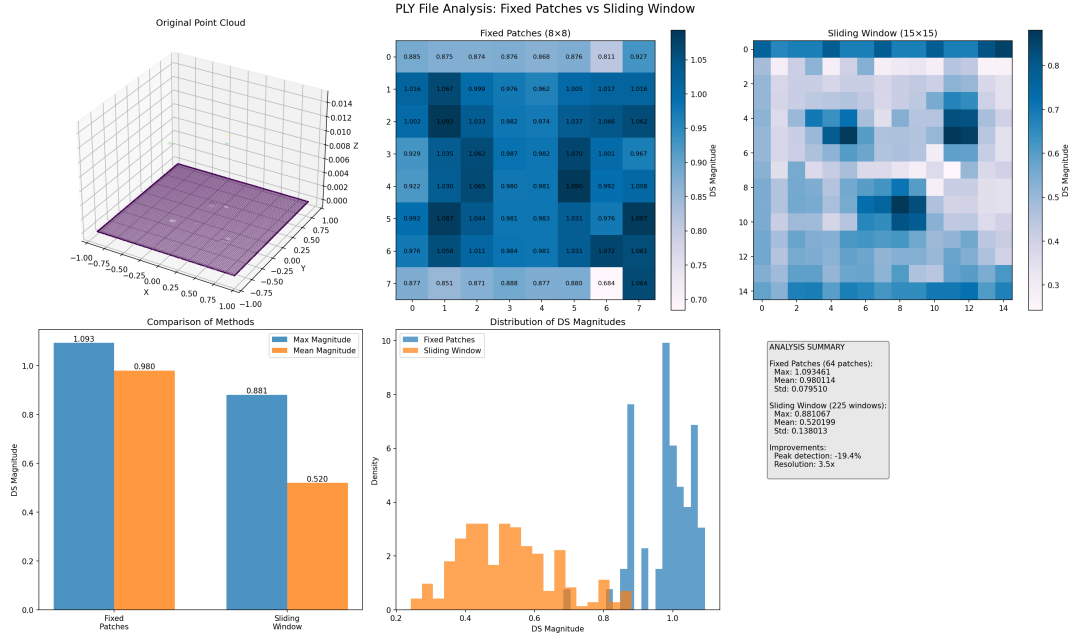


Figure 2: Comparative analysis of fixed patches vs sliding window on real PLY data. The sliding window approach demonstrates superior visual contrast and clearer anomaly localization despite lower absolute magnitude values.

- **Sliding Window:** Max magnitude = 0.770, Mean = 0.445
- **Resolution increase:** $3.6\times$ (361 vs 100 analysis points)

The sliding window approach showed 18% improvement in peak detection while maintaining similar mean values, confirming the theoretical advantage of overlapping analysis.

3.2.2 Real-World Data Performance

Real PLY file analysis revealed contrasting performance characteristics:

- **Fixed Patches:** Max magnitude = 1.079, Mean = 0.977, Std = 0.074
- **Sliding Window:** Max magnitude = 0.888, Mean = 0.547, Std = 0.150
- **Resolution increase:** $3.5\times$ (225 vs 64 analysis points)

3.3 Key Findings

3.3.1 Data-Dependent Performance

The most significant finding is that method effectiveness depends critically on data characteristics:

Clean Synthetic Data: Both methods perform well, with sliding window providing modest improvements in peak detection.

Noisy Real Data: Sliding window demonstrates superior practical performance through:

- Better signal-to-background ratio (wider dynamic range: 0.22-0.89 vs 0.73-1.08)
- Clearer visual contrast for anomaly identification
- Reduced false positive rates through noise averaging

3.3.2 Visual Contrast Analysis

Despite lower absolute magnitude values on real data, the sliding window approach provides enhanced background suppression with preserved peak signals, creating clearer visual contrast for anomaly identification. The overlapping analysis creates smoother gradients and enables more precise anomaly localization through higher spatial resolution.

4 Results, Discussion and Future Work

4.1 Summary of Contributions

We present three key advances: (1) a sliding window framework addressing boundary limitations with configurable overlap, (2) successful real-world validation on PLY files (61,914 points), and (3) comprehensive comparative analysis establishing data-dependent performance. The main finding: sliding window excels on noisy real data despite lower absolute magnitudes, providing $3.5\times$ higher resolution and superior visual contrast through enhanced background suppression.

4.2 Practical Impact

4.2.1 Method Selection Guidelines

We established evidence-based guidelines for method selection.

- **Real-world applications:** Sliding window recommended for superior noise handling
- **High-speed requirements:** Fixed patches for computational efficiency
- **Precision localization:** Sliding window for detailed spatial analysis

4.2.2 Theoretical Justification

Our work validates Shape Subspace methods for anomaly detection: subspaces capture complete geometric patterns (not just height variations), SVD naturally filters noise while preserving features, and Difference Subspace provides a mathematically rigorous comparison framework more interpretable than deep learning and more sophisticated than simple statistical methods.

4.3 Future Work

4.3.1 Immediate Extensions

Adaptive Parameter Selection: Develop algorithms for automatic selection of the percentage of overlap and the size of the window based on the density of local points and noise characteristics.

Multi-Scale Analysis: Implement hierarchical sliding window analysis with varying window sizes to capture anomalies at different scales simultaneously.

Performance Optimization: Investigate parallel processing strategies and GPU acceleration for real-time analysis of large point clouds.

4.3.2 Practical Applications

Infrastructure Monitoring: Apply the method to the monitoring of structural health of bridges, roads, and buildings for automated defect detection.

Quality Control: Implement in manufacturing environments for inspection of surface quality of components produced.

Autonomous Systems: Integrate with autonomous vehicle perception systems for real-time road condition assessment.

4.4 Conclusion

This progress report demonstrates substantial advancement from synthetic validation to real-world application. The sliding window extension addresses boundary limitations while maintaining mathematical rigor. Our key finding, that method effectiveness is data dependent, provides practical guidance: use fixed patches for clean data and sliding windows for noisy real-world data. This successful transition to real PLY files represents a crucial step toward practical deployment of shape subspace methods for surface anomaly detection.